PYTHON ACTIVITY BASED LEARNING

PROBLEM-7

Integrated University Fee and Eligibility Audit: Combine fee payments, attendance, internal marks, and examination eligibility records; identify students with outstanding conditions and generate a reason-wise report

Objective:

To develop an Integrated University Fee and Eligibility Audit System that combines student fee payment records, attendance percentages, internal marks, and examination eligibility details to identify students with outstanding dues or unmet eligibility requirements and generate a reason-wise report for accurate monitoring and decision-making.

Algorithm:

1. Start.

2.Input the number of students.

3.For each student, input name, total fee, paid fee, attendance percentage, internal marks, and examination approval status.

4.Input the minimum attendance percentage and minimum internal marks required.

5.Initialize all reason-wise counters to zero.

6.For each student:

- Calculate outstanding fee as total fee minus paid fee.
- Check whether any fee is outstanding.
- Check whether attendance is below the minimum requirement.
- Check whether internal marks are below the minimum requirement.
- Check whether examination eligibility is approved.

7.Store all applicable reasons for each student.

8.Display ALL CLEAR if no outstanding conditions exist; otherwise, display OUTSTANDING and list the reasons.

9.Update the corresponding counters for each outstanding condition.

10.Display the reason-wise audit summary.

11. Stop.

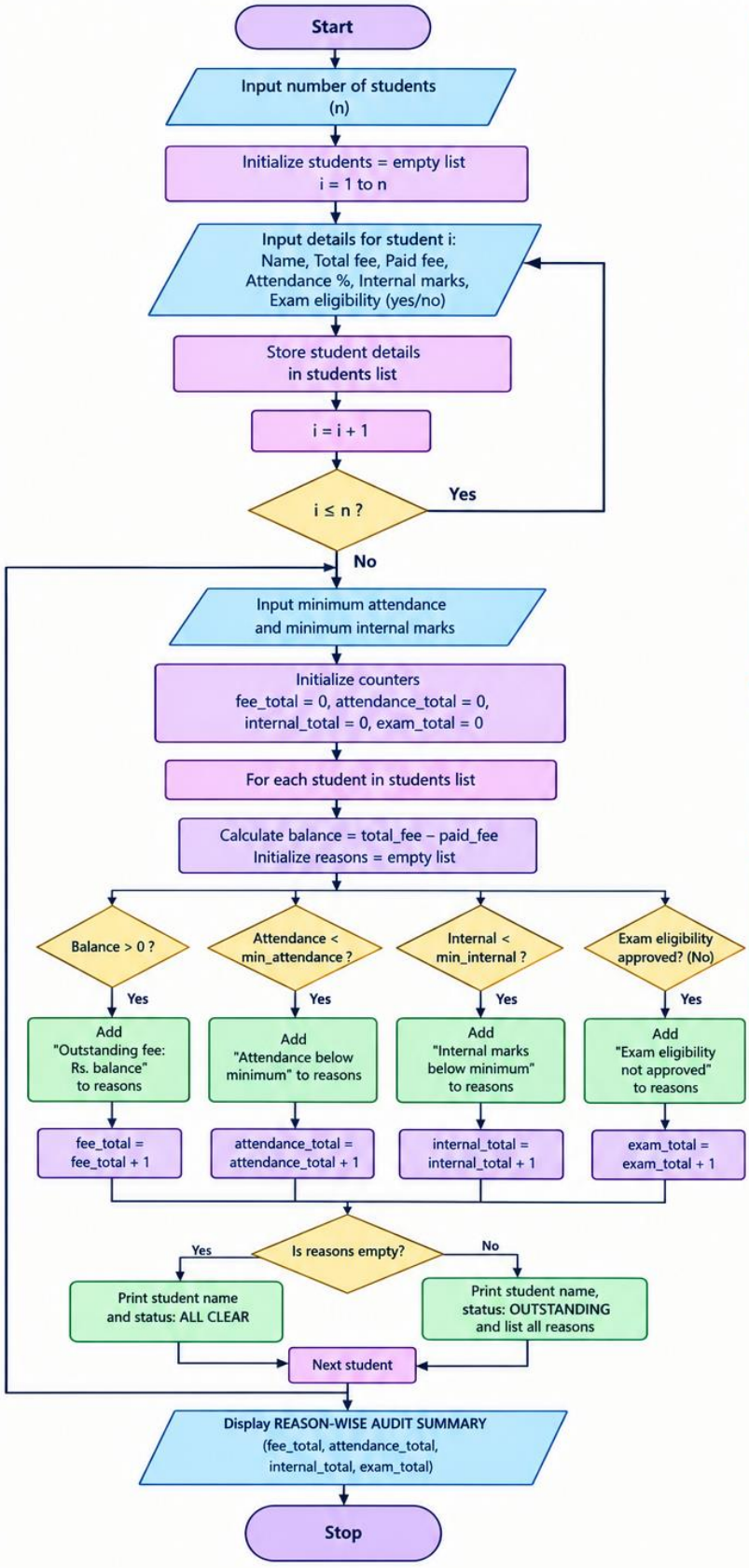
Pseudocode:

```
BEGIN
INPUT number_of_students
INITIALIZE students = empty list
FOR each student
    INPUT name
    INPUT total_fee
    INPUT paid_fee
    INPUT attendance
    INPUT internal_marks
    INPUT exam_approval
    STORE student details
END FOR
INPUT min_attendance
INPUT min_internal
INITIALIZE fee_total = 0
INITIALIZE attendance_total = 0
INITIALIZE internal_total = 0
INITIALIZE exam_total = 0
FOR each student in students
    INITIALIZE reasons = empty list
    balance = total_fee - paid_fee
    IF balance > 0 THEN
        ADD outstanding fee reason
        fee_total = fee_total + 1
    END IF
    IF attendance < min_attendance THEN
        ADD low attendance reason
```

```
        attendance_total = attendance_total + 1
    END IF
    IF internal_marks < min_internal THEN
        ADD low internal marks reason
        internal_total = internal_total + 1
    END IF
    IF exam_approval is FALSE THEN
        ADD exam approval reason
        exam_total = exam_total + 1
    END IF
    IF reasons is empty THEN
        DISPLAY student name, "ALL CLEAR"
    ELSE
        DISPLAY student name, "OUTSTANDING"
        DISPLAY all reasons
    END IF
END FOR
DISPLAY "REASON-WISE AUDIT SUMMARY"
DISPLAY fee_total
DISPLAY attendance_total
DISPLAY internal_total
DISPLAY exam_total
STOP
```

Flowchart:

Integrated University Fee and Eligibility Audit
(Flow Chart)



Input Details

- Number of students
- For each student:
  - Name
  - Total fee
  - Paid fee
  - Attendance percentage
  - Internal marks
  - Exam eligibility (yes/no)
- Minimum attendance percentage
- Minimum internal marks

Eligibility Conditions

- **Fee:** No outstanding amount
- **Attendance:** ≥ minimum attendance
- **Internal marks:** ≥ minimum internal
- **Exam eligibility:** Approved

Ranking / Audit Output

- For each student:
  - ALL CLEAR (if no outstanding)
  - OUTSTANDING (with reasons)

Reason-wise Summary

- Students with outstanding fees
- Students with low attendance
- Students with low internal marks
- Students without exam approval

Key Variables

- fee_total
- attendance_total
- internal_total
- exam_total

Python Program:

```
# Integrated University Fee and Eligibility Audit

n = int(input("Enter number of students: "))

students = []

for i in range(n):
    print("\nEnter details for Student", i + 1)

    name = input("Enter student name: ")
    total_fee = float(input("Enter total fee: "))
    paid_fee = float(input("Enter fee paid: "))
    attendance = float(input("Enter attendance percentage: "))
    internal = float(input("Enter internal marks: "))
    exam_input = input("Exam eligibility approved? (yes/no): ")

    exam_ok = exam_input.lower() == "yes"

    students.append([
        name, total_fee, paid_fee,
        attendance, internal, exam_ok
    ])

min_attendance = float(input("\nEnter minimum attendance: "))
min_internal = float(input("Enter minimum internal marks: "))
```

```
fee_total = 0
attendance_total = 0
internal_total = 0
exam_total = 0

print("\nUNIVERSITY FEE AND ELIGIBILITY AUDIT")
print("=" * 50)

for student in students:
    name, total_fee, paid_fee, attendance, internal, exam_ok = student

    reasons = []
    balance = total_fee - paid_fee

    # Fee audit
    if balance > 0:
        reasons.append("Outstanding fee: Rs." + str(round(balance, 2)))
        fee_total += 1

    # Attendance audit
    if attendance < min_attendance:
        reasons.append("Attendance below minimum")
        attendance_total += 1

    # Internal marks audit
    if internal < min_internal:
        reasons.append("Internal marks below minimum")
        internal_total += 1
```

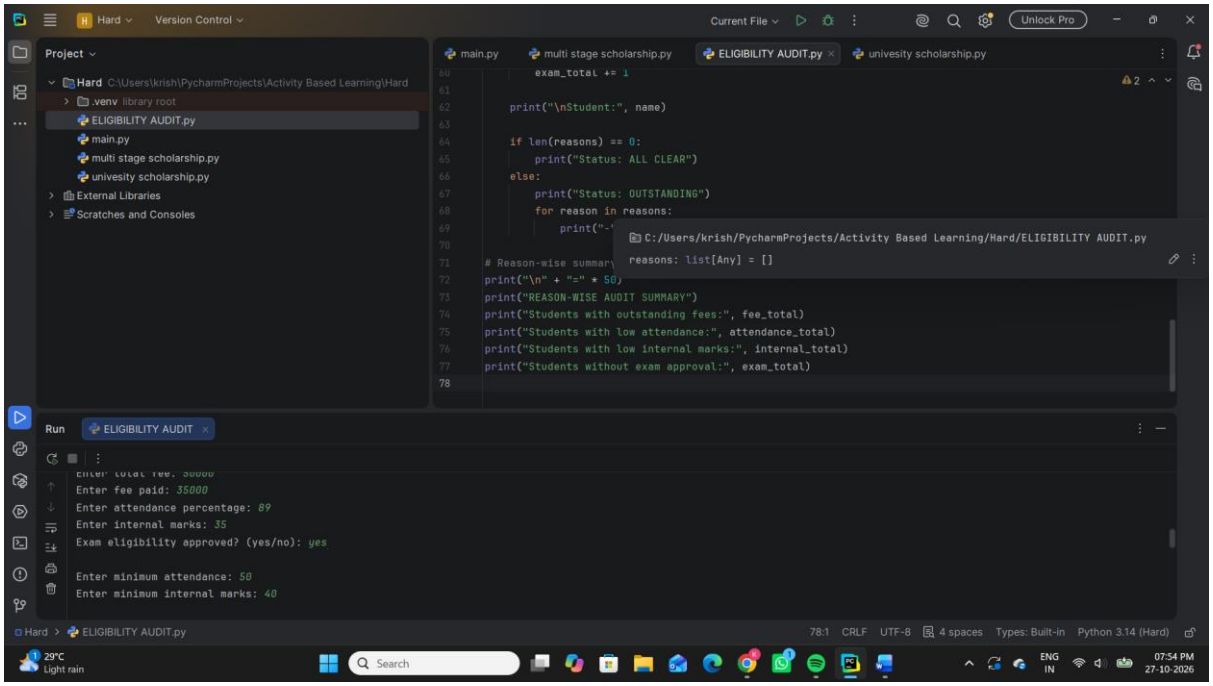
```
    # Examination eligibility audit
    if not exam_ok:
        reasons.append("Exam eligibility not approved")
        exam_total += 1

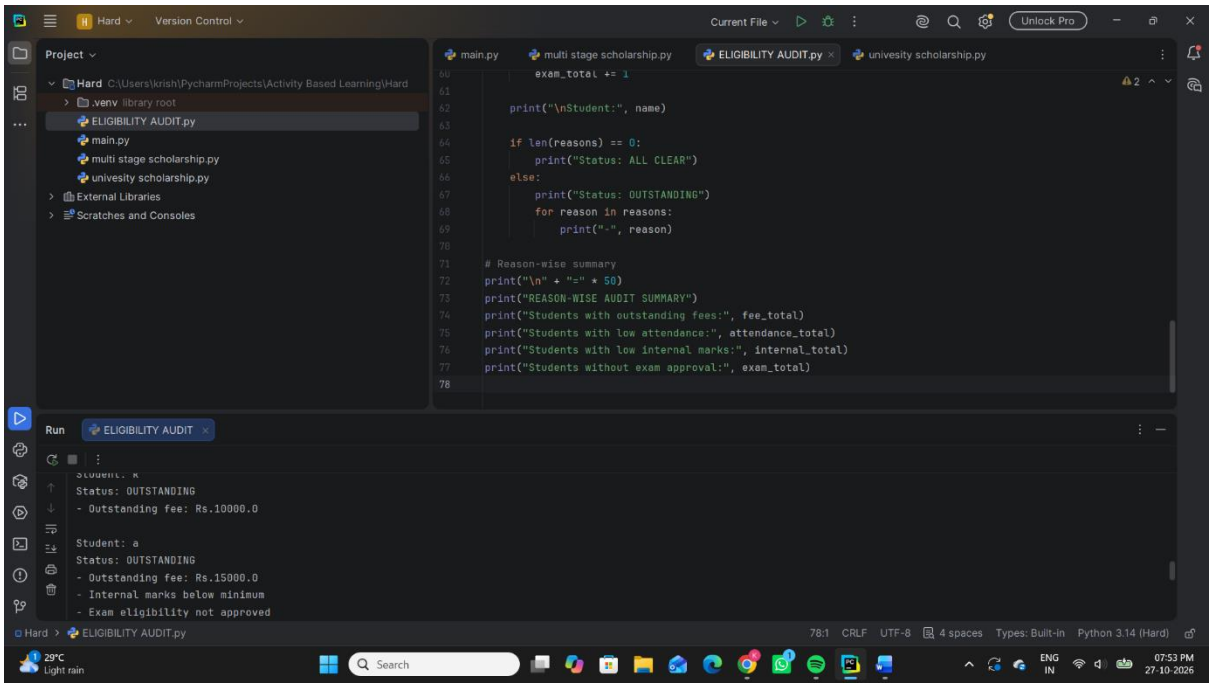
    print("\nStudent:", name)

    if len(reasons) == 0:
        print("Status: ALL CLEAR")
    else:
        print("Status: OUTSTANDING")
        for reason in reasons:
            print("-", reason)

# Reason-wise summary
print("\n" + "=" * 50)
print("REASON-WISE AUDIT SUMMARY")
print("Students with outstanding fees:", fee_total)
print("Students with low attendance:", attendance_total)
print("Students with low internal marks:", internal_total)
print("Students without exam approval:", exam_total)
```

Output Screen Shots:


Hard
Version Control
Current File
Unlock Pro
Project
Hard C:\Users\krish\PycharmProjects\Activity Based Learning\Hard
.venv library root
ELIGIBILITY AUDIT.py
main.py
multi stage scholarship.py
univesity scholarship.py
External Libraries
Scratches and Consoles
main.py
multi stage scholarship.py
ELIGIBILITY AUDIT.py
univesity scholarship.py
exam_total += 1
print("\nStudent:", name)
if len(reasons) == 0:
print("Status: ALL CLEAR")
else:
print("Status: OUTSTANDING")
for reason in reasons:
print("-"
C:/Users/krish/PycharmProjects/Activity Based Learning/Hard/ELIGIBILITY AUDIT.py
reasons: list[Any] = []
Reason-wise summary
print("\n" + "=" * 50)
print("REASON-WISE AUDIT SUMMARY")
print("Students with outstanding fees:", fee_total)
print("Students with low attendance:", attendance_total)
print("Students with low internal marks:", internal_total)
print("Students without exam approval:", exam_total)
Run
ELIGIBILITY AUDIT
Enter fee paid: 35000
Enter attendance percentage: 89
Enter internal marks: 35
Exam eligibility approved? (yes/no): yes
Enter minimum attendance: 50
Enter minimum internal marks: 40
Hard
ELIGIBILITY AUDIT.py
78:1
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Hard)
29°C
Light rain
Search
ENG
IN
07:54 PM
27-10-2026


Hard
Version Control
Current File
Unlock Pro
Project
Hard C:\Users\krish\PycharmProjects\Activity Based Learning\Hard
.venv library root
ELIGIBILITY AUDIT.py
main.py
multi stage scholarship.py
univesity scholarship.py
External Libraries
Scratches and Consoles
main.py
multi stage scholarship.py
ELIGIBILITY AUDIT.py
univesity scholarship.py
exam_total += 1
print("\nStudent:", name)
if len(reasons) == 0:
print("Status: ALL CLEAR")
else:
print("Status: OUTSTANDING")
for reason in reasons:
print("-", reason)
Reason-wise summary
print("\n" + "=" * 50)
print("REASON-WISE AUDIT SUMMARY")
print("Students with outstanding fees:", fee_total)
print("Students with low attendance:", attendance_total)
print("Students with low internal marks:", internal_total)
print("Students without exam approval:", exam_total)
Run
ELIGIBILITY AUDIT
Status: OUTSTANDING
- Outstanding fee: Rs.10000.0
Student: a
Status: OUTSTANDING
- Outstanding fee: Rs.15000.0
- Internal marks below minimum
- Exam eligibility not approved
Hard
ELIGIBILITY AUDIT.py
78:1
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Hard)
29°C
Light rain
Search
ENG
IN
07:53 PM
27-10-2026